W-R-36c
November 7, 1963

SUPERSEDING
Int. Fed. Spec. W-R-0036b (GSA-FSS)
January 11, 1963 and
Fed. Spec. W-R-36a
January 29, 1957

FEDERAL SPECIFICATION

RACEWAYS AND FITTINGS, METALLIC, UNDERFLOOR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers metallic underfloor duct systems designed for use as raceways for the installation of wires and cables.

1.2 Classification.

1.2.1 *Types.* Underfloor metallic raceway furnished under this specification shall be of the following types, as specified (see 6.1).

Type I.—Covered.

Type II.—Flush.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-P-116—Preservation, Methods of.

Military Standards:

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5975

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2.2 Other publications. The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification.

Underwriters' Laboratories, Inc., Standard:

UL 884 — Standard for Underfloor Raceways and Fittings.

(Copies may be obtained from Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N. Y. 11749; 207 E. Ohio Street, Chicago, Illinois 60611; and 1655 Scott Blvd., Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 Except as otherwise provided herein, metallic raceways and fittings covered by this specification shall conform to the applicable requirements of Underwriters' Laboratories, Inc., Standard UL-884 for Underfloor Raceways and Fittings. All boxes, couplings, elbows, bends, duct supports, etc., shall be suitable for use with the duct furnished to form the complete system.

3.2 Fire and casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency proof that underfloor metallic raceways and fittings he proposes to supply under this specification conform to the requirements of UL-884. The label, or listing with re-examination, of the Underwriters' Laboratories, Inc., may be accepted as evidence that the underfloor metallic raceways and fittings conform to this requirement.

3.2.2 In lieu of the label, or listing with re-examination, the bidder may submit independent proof, satisfactory to the contracting agency, that underfloor metallic raceways and fittings conform to the applicable requirements of the published standards, including methods of tests, of the Underwriters' Laboratories, Inc., UL-884 Standard.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.3 Raceways.

3.3.1 *Dimensions.* The minimum external height of raceways shall be 1-1/4 inches and the minimum internal cross-sectional area shall be three square inches. The tolerance on the external dimensions shall not exceed 1.5 percent of the minimum dimension. The standard length of a raceway section shall be 10 feet, plus 0 minus 1/4 inch. Where lengths shorter than 10 feet are required, the tolerances shall be reduced in proportion to the length ratio. For lengths greater than 10 feet, the tolerances shall not exceed those for the standard length.

3.3.2 *Insert or plug openings.* Except where blank duct is specified, each length of raceway shall be provided with suitable inserts or plugs. The connection between the insert and outlet fitting shall employ not less than three full threads. Standard spacing shall be at 2-foot intervals (five inserts or plugs per ten-foot length) unless special spacing is specified. The size of standard pipe threads for inserts and plugs shall be 2-inch for type I duct and 1-1/4-inch for type II duct.

3.3.3 *Straightness.* When tested as specified in 4.3.1.1, the gap in a 10-foot length of raceway shall not exceed 1/8 inch in the horizontal plane and 1/4 inch in the vertical plane.

3.3.4 *Twisting.* When tested as specified in 4.3.1.2, the height of either corner shall not exceed 3/16 inch.

3.4 Inserts and plugs.

3.4.1 *Height.* The nominal height of inserts shall be as specified (see 6.1). The actual height of the insert will be 1/8 inch less

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than the nominal height to accommodate a layer of concrete approximately 1/8 inch thick over the capped insert when installed in the system. Plugs for type II duct shall finish flush with the top of the duct.

3.5 Junction boxes.

3.5.1 *Tunnels.* Where a junction box is provided with a conduit corner entrance, the cross-sectional area of the conduit tunnels shall be not less than 100 percent of the cross-sectional area of the rigid metal conduit which is to be connected to the junction box.

3.5.2 *Cover plate deflection.* When tested as specified in 4.3.2.1, the permanent deflection of the cover plate, exclusive of the gasket depression, shall be not more than 3/64 inch.

3.5.2.1 A holder for tile, concrete, or terrazzo, or a flush plate shall be considered as finishing material and need not meet the deflection requirement unless it is a permanent part of the cover plate.

3.5.3 *Duct openings.* Duct openings on multiple duct boxes shall provide a spacing of 1/2 inch minimum between ducts.

3.6 Fittings and accessories.

3.6.1 *Outlet fittings (power, telephone, etc.)*

3.6.1.1 *Height.* Outlet fittings shall be of minimum height but shall not exceed 3 inches unless greater height is necessitated by electrical clearance within the fitting.

3.6.1.2 *Fasteners.* Assembly and disassembly of the outlet housing shall be readily accomplished. Receptacle assembly screws shall not be self-tapping.

3.6.2 *Standpipe fittings.*

3.6.2.1 *Size.* Standpipe fittings shall be 3/4, 1, 1-1/4, 1-1/2, and 2 inches standard pipe size, as specified (see 6.1), with suitable adapter to fit the duct outlet.

3.6.2.2 *Openings.* To prevent injuries to wires or cables, all openings of standpipes and adapters through which cables pass shall be bushed or finished with well-rounded inside edges.

3.6.2.3 *Length.* Unless otherwise specified, standpipe lengths shall be 3 inches.

3.6.3 *Recessed floor fittings.* Recessed floor fittings shall finish approximately flush with the finished floor or have flanges that cover any unused opening in the floor finish. This fitting shall be adjustable to the floor finish.

3.6.3.1 *Attachments.* Means for attaching kick posts, bell nozzles, strain reliefs, and abandoning plates shall be provided. These means of attaching shall provide a tight seal and be mechanically strong. Fittings shall enclose receptacles and terminals except for wiring accesses.

3.7 Identification marking (military agencies). Raceways and fittings shall be marked for identification in accordance with MIL-STD-130.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Except as otherwise provided herein, sampling, inspection, and tests shall be in conformity with the requirements therefor specified and described in Underwriters' Laboratories, Inc. UL-884.

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4.3 Tests. In addition to the tests specified in UL-884, raceways and fittings shall be subjected to the following tests, as applicable.

4.3.1 *Raceways.*

4.3.1.1 *Straightness.* Straightness shall be determined by placing a 10-foot length of raceway against a straightedge and measuring the gap between the raceway and the straightedge.

4.3.1.2 *Twisting.* Twist shall be determined by holding the edge of one end of a 10-foot length of raceway down on a surface plate, with the bottom side of the duct parallel to the surface plate, and measuring the height of either corner of the opposite end above the surface plate.

4.3.2 *Junction boxes.*

4.3.2.1 *Cover plate deflection.* Plate deflection shall be determined in the following manner. A junction box, with cover plate in place, shall be used. After raising all adjusting screws 1/8 inch, a load of 1500 pounds shall be applied to the center of the cover plate for one minute by means of a 3-inch mandrel with no sharp edges. The deflection during the time the load is applied shall be not more than 1/8 inch.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, Fed. Std. No. 102 should be referred to for the definitions and applications of the various levels of packaging and packing.)

5.1 Packaging.

5.1.1 *Level A.* The raceways and fittings shall be packaged by method III of MIL-P-116.

5.1.2 *Level C.* The raceways and fittings shall be preserved and packaged in accordance with the manufacturer's practice.

5.2 Packing.

5.2.1 *Level* A. The raceways and fittings shall be packed in overseas type cleated-plywood boxes conforming to PPP-B-601, or overseas class nailed wood boxes conforming to PPP-B-621. The gross weight of boxes shall not exceed approximately 200 pounds.

5.2.2 *Level B.* The raceways and fittings shall be packed in domestic type cleated plywood boxes conforming to PPP-B-601, domestic class nailed wood boxes conforming to PPP-B-621, or exterior grade fiberboard boxes conforming to PPP-B-636. Contents of boxes shall be cushioned and braced to prevent movement within the boxes. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

5.2.3 *Level C.* The raceways and fittings shall be packed or prepared for shipment in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.

(*b*) Type of raceway, (1.2.1).

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(*c*) Length of raceway, if other than standard (3.3.1).

(*d*) Internal cross-sectional area of raceway, if significant.

(*e*) Insert or plug spacing, if other than specified (3.3.2).

(*f*) Height of inserts required (3.4.1).

(*g*) Size of standpipe fittings required (3.6.2.1)

(*h*) Length of standpipe, if other than specified (3.6.2.3).

(*i*) Level of packaging and packing required (5.1 and 5.2).

(*j*) Special marking, if required (5.3).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Conductor pipe, steel.

Carload minimum weight 20,000 pounds, subject to Rule 34 Uniform Freight Classification.

Motor:

Conductor pipe, steel.

Truckload minimum weight 20,000 pounds, subject to Rule 115, National Motor Freight Classification

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MILITARY INTEREST:

Army—C E

Navy—Y

Air Force

Copies of this specification may be purchased fo cents each.

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